

⚙️ Cloud Data Lake Status

🔄 Sync & Ingest from AWS S3

Data Inventory Tracking

📁 products.csv

8 Rows

📁 customers.csv

10,000 Rows

📁 sales.csv

500,000 Rows



Retail Operations Intelligence Engine

Production-Grade Agentic Big Data Pipeline with LangGraph & MLflow



Query the Operations Analyst



Try a sample operational query:

Find the top 3 spending customers and tell me which cities they are from



Or enter a custom business analyst question:

Find the top 3 spending customers and tell me which cities they are from



Execute Agent Analysis



Analytical Insights

Executive Summary: Top Revenue-Generating Customers

Overview An analysis of account spending identifies our top three individual customers by total revenue, highlighting significant revenue concentration in Ch

Key Findings

Rank	Customer ID	Customer Name	Location	Total Revenue
1	CUST-12206	Meera Mehta	Chennai	\$22,849.59
2	CUST-15954	Kabir Mehta	Chennai	\$22,458.06
3	CUST-19176	Meera Singh	Delhi	\$22,165.84

Strategic Takeaways

- **Geographic Focus (Chennai Market Power):** 67% of top-tier spenders (2 out of 3) are located in Chennai, indicating strong brand loyalty and higher client concentration in the South.
- **Revenue Parity:** Spending among top clients is tightly clustered within a narrow margin (~\$680 variance between #1 and #3), pointing to a consistent purchasing power across the top tier.
- **Action Item:** Recommend establishing targeted VIP account management programs in **Chennai** and **Delhi** to preserve client retention and leverage local market insights.

Self-Correction Retries Required

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✂ Execution Code Trace

Executed Pandas Data Manipulation Pipeline Code:

```
import pandas as pd

try:
    sales
except NameError:
    sales = pd.read_csv("sales.csv")

try:
    customers
except NameError:
    customers = pd.read_csv("customers.csv")

# Calculate total spending per customer
customer_spending = (
    sales.groupby("Customer_ID")["Total_Revenue"].sum().reset_index()
)

# Get top 3 spending customers
top_3_spending = customer_spending.sort_values(
    by="Total_Revenue", ascending=False
).head(3)

# Merge with customers dataframe to include Customer_Name and City
result = top_3_spending.merge(customers, on="Customer_ID")[
    ["Customer_ID", "Customer_Name", "City", "Total_Revenue"]
]
```

Calculated Raw Context Outputs:

	Customer_ID	Customer_Name	City	Total_Revenue
0	CUST-12206	Meera Mehta	Chennai	22,849.59
1	CUST-15954	Kabir Mehta	Chennai	22,458.06
2	CUST-19176	Meera Singh	Delhi	22,165.84

 View Live System Graph Architecture

